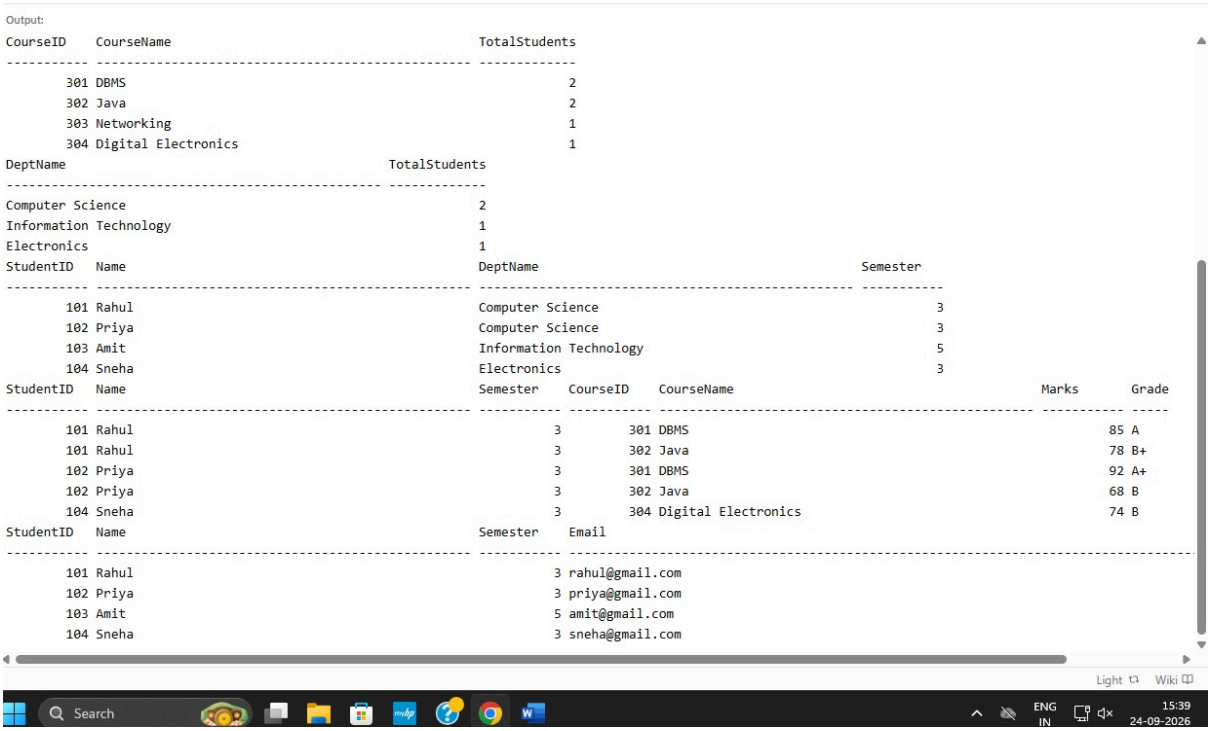
```
-- 1. CREATE TABLES

CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50),
    HOD VARCHAR(50),
    Building VARCHAR(50)
);
GO

CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
    DOB DATE,
    DeptID INT,
    Semester INT,
    Email VARCHAR(100),
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);
GO
```

STDIN

Input for the program (Optional)



Output:

```
CourseID    CourseName                                    TotalStudents
----------- --------------------------------------------- -------------
       301 DBMS                                                      2
       302 Java                                                      2
       303 Networking                                                1
       304 Digital Electronics                                       1
DeptName                                          TotalStudents
------------------------------------------------- -------------
Computer Science                                  2
Information Technology                            1
Electronics                                       1
StudentID   Name                                  DeptName                                          Semester
----------- --------------------------------------------- ------------------------------------------------- -----------
       101 Rahul                                 Computer Science                                          3
       102 Priya                                 Computer Science                                          3
       103 Amit                                  Information Technology                                    5
       104 Sneha                                 Electronics                                               3
StudentID   Name                                  Semester    CourseID    CourseName                                        Marks       Grade
----------- --------------------------------------------- ----------- ----------- ------------------------------------------------- ----------- -----
       101 Rahul                                           3         301 DBMS                                              85 A
       101 Rahul                                           3         302 Java                                              78 B+
       102 Priya                                           3         301 DBMS                                              92 A+
       102 Priya                                           3         302 Java                                              68 B
       104 Sneha                                           3         304 Digital Electronics                               74 B
StudentID   Name                                  Semester    Email
----------- --------------------------------------------- ----------- ----------------------------------------------------------------------------------------------------
       101 Rahul                                           3 rahul@gmail.com
       102 Priya                                           3 priya@gmail.com
       103 Amit                                            5 amit@gmail.com
       104 Sneha                                           3 sneha@gmail.com
```

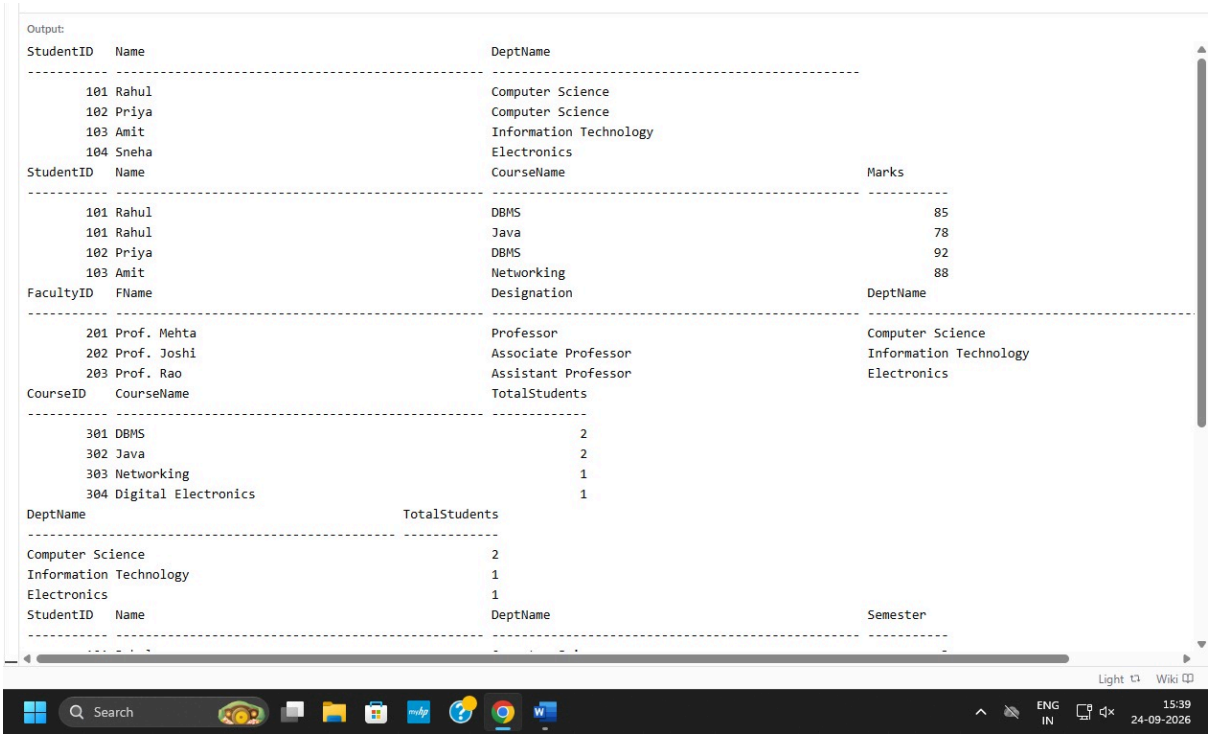
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```
CREATE TABLE Faculty (
    FacultyID INT PRIMARY KEY,
    FName VARCHAR(50),
    DeptID INT,
    Designation VARCHAR(50),
    Salary DECIMAL(10,2),
```

```
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);
GO

CREATE TABLE Course (
    CourseID INT PRIMARY KEY,
    CourseName VARCHAR(50),
    DeptID INT,
    Credits INT,
    FacultyID INT,
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID),
    FOREIGN KEY (FacultyID) REFERENCES Faculty(FacultyID)
);
GO
```



Output:

```
StudentID   Name                                               DeptName
----------- -------------------------------------------------- --------------------------------------------------
        101 Rahul                                              Computer Science
        102 Priya                                              Computer Science
        103 Amit                                               Information Technology
        104 Sneha                                              Electronics
StudentID   Name                                               CourseName                                         Marks
----------- -------------------------------------------------- -------------------------------------------------- -----------
        101 Rahul                                              DBMS                                                    85
        101 Rahul                                              Java                                                    78
        102 Priya                                              DBMS                                                    92
        103 Amit                                               Networking                                              88
FacultyID   FName                                              Designation                                        DeptName
----------- -------------------------------------------------- -------------------------------------------------- --------------------------------------------------
        201 Prof. Mehta                                        Professor                                          Computer Science
        202 Prof. Joshi                                        Associate Professor                                Information Technology
        203 Prof. Rao                                          Assistant Professor                                Electronics
CourseID    CourseName                                         TotalStudents
----------- -------------------------------------------------- -------------
        301 DBMS                                                           2
        302 Java                                                           2
        303 Networking                                                     1
        304 Digital Electronics                                            1
DeptName                                           TotalStudents
-------------------------------------------------- -------------
Computer Science                                   2
Information Technology                             1
Electronics                                        1
StudentID   Name                                               DeptName                                           Semester
----------- -------------------------------------------------- -------------------------------------------------- -----------
```



```
CREATE TABLE Enrollment (
    EnrollID INT PRIMARY KEY,
    StudentID INT,
    CourseID INT,
    Semester INT,
    Marks INT,
    Grade VARCHAR(5),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);
GO

-- INSERT DATA
```

```
GO

INSERT INTO Department VALUES
(1, 'Computer Science', 'Dr. Sharma', 'Block A'),
(2, 'Information Technology', 'Dr. Patil', 'Block B'),
(3, 'Electronics', 'Dr. Kumar', 'Block C');
GO

INSERT INTO Student VALUES
(101, 'Rahul', '2004-05-10', 1, 3, 'rahul@gmail.com'),
(102, 'Priya', '2004-08-15', 1, 3, 'priya@gmail.com'),
(103, 'Amit', '2003-12-20', 2, 5, 'amit@gmail.com'),
(104, 'Sneha', '2004-03-25', 3, 3, 'sneha@gmail.com');
GO

INSERT INTO Faculty VALUES
(201, 'Prof. Mehta', 1, 'Professor', 75000),
(202, 'Prof. Joshi', 2, 'Associate Professor', 65000),
(203, 'Prof. Rao', 3, 'Assistant Professor', 55000);
GO

INSERT INTO Course VALUES
(301, 'DBMS', 1, 4, 201),
(302, 'Java', 1, 4, 201),
(303, 'Networking', 2, 3, 202),
(304, 'Digital Electronics', 3, 4, 203);
GO

INSERT INTO Enrollment VALUES
(401, 101, 301, 3, 85, 'A'),
(402, 101, 302, 3, 78, 'B+'),
(403, 102, 301, 3, 92, 'A+'),
(404, 102, 302, 3, 68, 'B'),
(405, 103, 303, 5, 88, 'A'),
(406, 104, 304, 3, 74, 'B');
GO

-- QUESTION 1
-- StudentInfo View

CREATE VIEW StudentInfo AS
SELECT
   s.StudentID,
   s.Name,
   d.DeptName
FROM Student s
INNER JOIN Department d
   ON s.DeptID = d.DeptID;
```

```
GO

SELECT * FROM StudentInfo;
GO

-- QUESTION 2
-- TopScorers View

CREATE VIEW TopScorers AS
SELECT
   s.StudentID,
   s.Name,
   c.CourseName,
   e.Marks
FROM Student s
INNER JOIN Enrollment e
   ON s.StudentID = e.StudentID
INNER JOIN Course c
   ON e.CourseID = c.CourseID
WHERE e.Marks > 75;
GO

SELECT * FROM TopScorers;
GO

-- QUESTION 3

CREATE VIEW FacultyDeptView AS
SELECT
   f.FacultyID,
   f.FName,
   f.Designation,
   d.DeptName
FROM Faculty f
INNER JOIN Department d
   ON f.DeptID = d.DeptID;
GO

SELECT * FROM FacultyDeptView;
GO

-- QUESTION 5

CREATE VIEW CourseEnrollmentCount AS
SELECT
   c.CourseID,
   c.CourseName,
   COUNT(e.StudentID) AS TotalStudents
```

```
FROM Course c
LEFT JOIN Enrollment e
    ON c.CourseID = e.CourseID
GROUP BY
    c.CourseID,
    c.CourseName;
GO

SELECT * FROM CourseEnrollmentCount;
GO

-- QUESTION 6
-- DepartmentStrength

CREATE VIEW DepartmentStrength AS
SELECT
    d.DeptName,
    COUNT(s.StudentID) AS TotalStudents
FROM Department d
LEFT JOIN Student s
    ON d.DeptID = s.DeptID
GROUP BY
    d.DeptID,
    d.DeptName;
GO

SELECT * FROM DepartmentStrength;
GO

-- ========================================
-- QUESTION 7
-- MODIFY StudentInfo VIEW
-- ========================================

ALTER VIEW StudentInfo AS
SELECT
    s.StudentID,
    s.Name,
    d.DeptName,
    s.Semester
FROM Student s
INNER JOIN Department d
    ON s.DeptID = d.DeptID;
GO

SELECT * FROM StudentInfo;
GO
```

```
-- ========================================
-- QUESTION 9
-- Semester III Students
-- ========================================

CREATE VIEW SemesterIIIStudents AS
SELECT
    s.StudentID,
    s.Name,
    s.Semester,
    c.CourseID,
    c.CourseName,
    e.Marks,
    e.Grade
FROM Student s
INNER JOIN Enrollment e
    ON s.StudentID = e.StudentID
INNER JOIN Course c
    ON e.CourseID = c.CourseID
WHERE s.Semester = 3;
GO

SELECT * FROM SemesterIIIStudents;
GO

-- ========================================
-- QUESTION 10
-- SIMPLE VIEW
-- ========================================

CREATE VIEW SimpleStudentView AS
SELECT
    StudentID,
    Name,
    Semester,
    Email
FROM Student;
GO

SELECT * FROM SimpleStudentView;
GO
```